Conformal Prediction for Uncertainty Quantification

Abstract

Traditional deep learning–based super-resolution (SR) [1] methods typically produce deterministic point estimates and lack any quantification of uncertainty in reconstructed MRI or Electromagnetic Imaging (EMI) images [2], a shortcoming that is particularly critical in clinical settings where high-confidence, but incorrect predictions can be misleading or harmful. To address this issue, this report presents a Conformal Prediction (CP)–based approach for assessing pixel-wise and region-wise uncertainty in SR reconstructions, delivering calibrated prediction intervals, clinically meaningful uncertainty maps, and reliability estimates for each SR output. In future work, these uncertainty measures will be directly incorporated into the SR training process through reliability-weighted loss functions, uncertainty-aware data sampling, and adaptive error correction, resulting in a more robust, interpretable, and clinically trustworthy modeling pipeline.

Methodology

A central challenge in super-resolution (SR) imaging is the lack of calibrated and reliable uncertainty quantification across multiple spatial scales, including image-level, pixel-level, and patch-level representations. This issue is further complicated by the inherently multi-domain nature of MRI and Electromagnetic Imaging (EMI) data. In MRI, spatial-domain representations capture smooth anatomical structures, whereas frequency-domain (k-space) representations emphasize edges and structural details. These domain-specific differences make uncertainty estimation particularly complex and often unstable when conventional calibration techniques are applied; similar challenges are observed in EMI data. To address these limitations, we propose a conformal prediction–based framework that provides statistically valid and well-calibrated uncertainty estimates for brain SR tasks. Due to limited availability of EMI data, the proposed methodology and evaluation are conducted using MRI data.

The proposed framework consists of four main stages. First, a lightweight convolutional neural network (CNN)–based SR model is trained using paired low-resolution (LR) and high-resolution (HR) MRI scans. The network directly predicts super-resolved images in the spatial domain from LR inputs. Second, a calibration stage is performed using a held-out dataset to compute nonconformity scores that quantify prediction errors. The nonconformity scores are stored and later used for quantile estimation. Third, conformal prediction is applied by sorting the nonconformity scores and estimating quantile thresholds corresponding to predefined uncertainty levels. These thresholds define the widths of the prediction intervals and provide formal statistical coverage guarantees under mild assumptions such as data exchangeability. Finally, during testing on unseen MRI scans, the trained SR model generates super-resolved outputs, and the calibrated quantile thresholds are applied to compute pixel-wise prediction intervals, resulting in uncertainty maps alongside the SR reconstructions and residual error maps.

The conformal framework is evaluated at multiple significance levels (α = 0.1, 0.05, 0.03). Smaller α values produce wider and more conservative prediction intervals, while larger α values yield tighter uncertainty bounds. The SR model architecture is a compact CNN optimized for computational efficiency and stability, consisting of three 3×3 convolutional layers, two ReLU activation functions, and a final regression output layer. The network operates directly in pixel space without normalization layers and processes 320×320 MRI slices as input. Training is performed using the mean squared error (MSE) loss between the predicted SR images and the corresponding high-resolution ground truth.

Calibration and Quantile Estimation

For each image in the calibration set, a nonconformity score is computed as the magnitude of the residual error between the super-resolved (SR) prediction and the corresponding high-resolution (HR) ground truth. These nonconformity scores quantify the model's prediction error and form the basis for uncertainty calibration. After aggregating all calibration scores, the conformal quantile q_α is estimated as the $(1-\alpha)$-quantile of the empirical nonconformity score distribution. This quantile defines the conformal bound that is subsequently applied at test time to construct statistically valid prediction intervals for the SR outputs.

The dataset used in the experiments (NINS-normal) [3] is partitioned into 1,694 images for training, 70 images for calibration, and one representative image used for qualitative visualization. On the test image, the proposed framework achieves a peak signal-to-noise ratio (PSNR) of approximately 26.75 dB and a structural similarity index (SSIM) of approximately 0.85, demonstrating competitive reconstruction quality while enabling calibrated uncertainty estimation.

Exchangeability Analysis

Conformal prediction fundamentally depends on the assumption of data exchangeability between the calibration and test samples. To validate this assumption, the proposed calibration procedure evaluates the distribution similarity between the calibration and test datasets by analyzing multiple statistical characteristics, including k-space magnitude distributions, spatial-domain residual statistics, and pixel-wise intensity histograms. These analyses ensure that the estimated conformal quantile thresholds remain valid and applicable when performing inference on previously unseen MRI scans.

To quantitatively assess exchangeability, a Kullback–Leibler (KL) divergence–based distribution [4] analysis is conducted between the calibration and test datasets. The resulting KL divergence distributions, shown in Fig. 1, demonstrate a high degree of similarity between the two sets, indicating consistency in their underlying data-generating processes. This empirical evidence supports the validity of the exchangeability assumption required by conformal prediction theory and confirms that the proposed framework adheres to its theoretical guarantees. Overall, the conformal prediction approach enables reliable and statistically calibrated uncertainty estimation for MRI super-resolution, offering a more transparent and clinically trustworthy reconstruction pipeline. The resulting uncertainty maps and prediction intervals improve interpretability by highlighting regions of reduced reconstruction confidence, thereby supporting more informed clinical decision-making. Fig. 2 illustrates the training performance of the proposed super-resolution network.

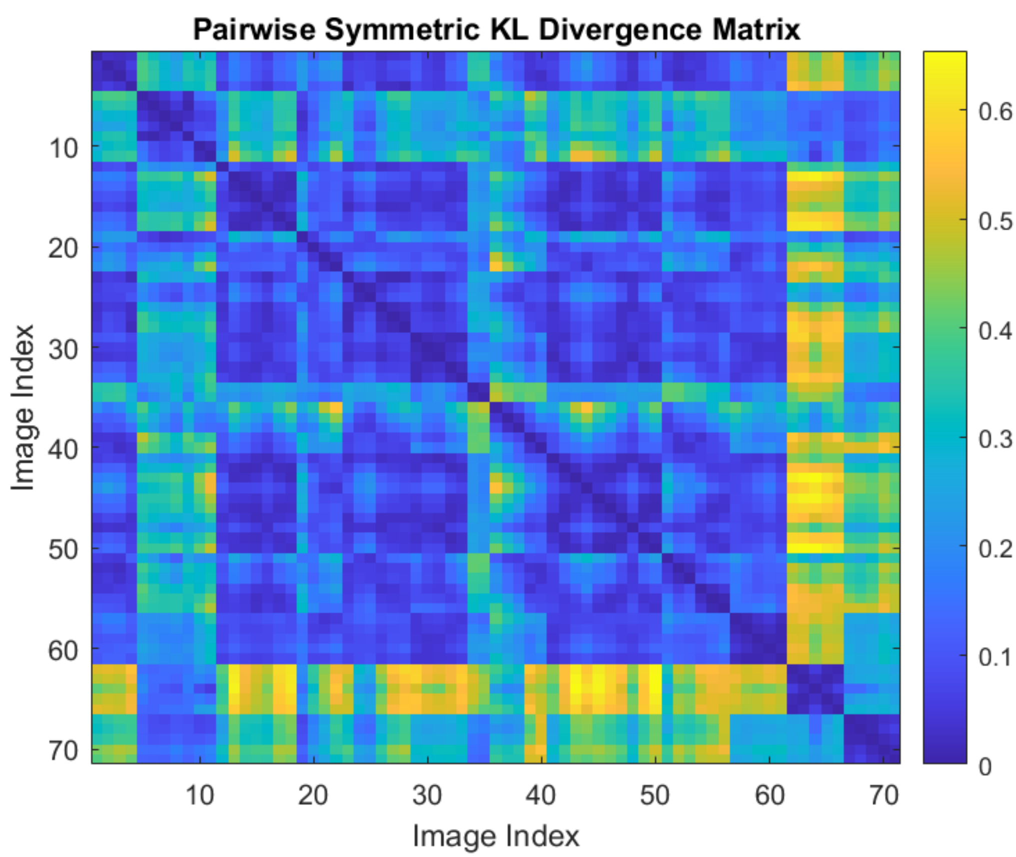

Fig. 1. The results of KL divergence analysis.

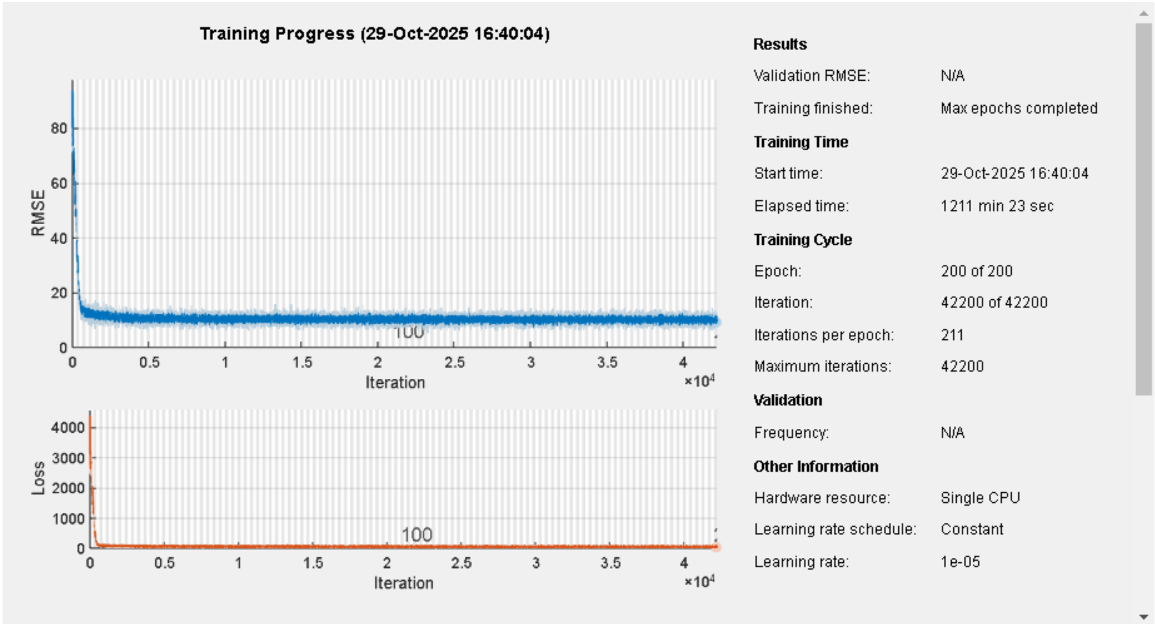

Fig. 2. The results obtained from training the proposed super-resolution network.

Figures 3, 4, and 5 illustrate the outcomes of applying conformal prediction with varying conformity thresholds to a single image sample. The results demonstrate how the choice of threshold influences the prediction regions and the model's uncertainty characterization.

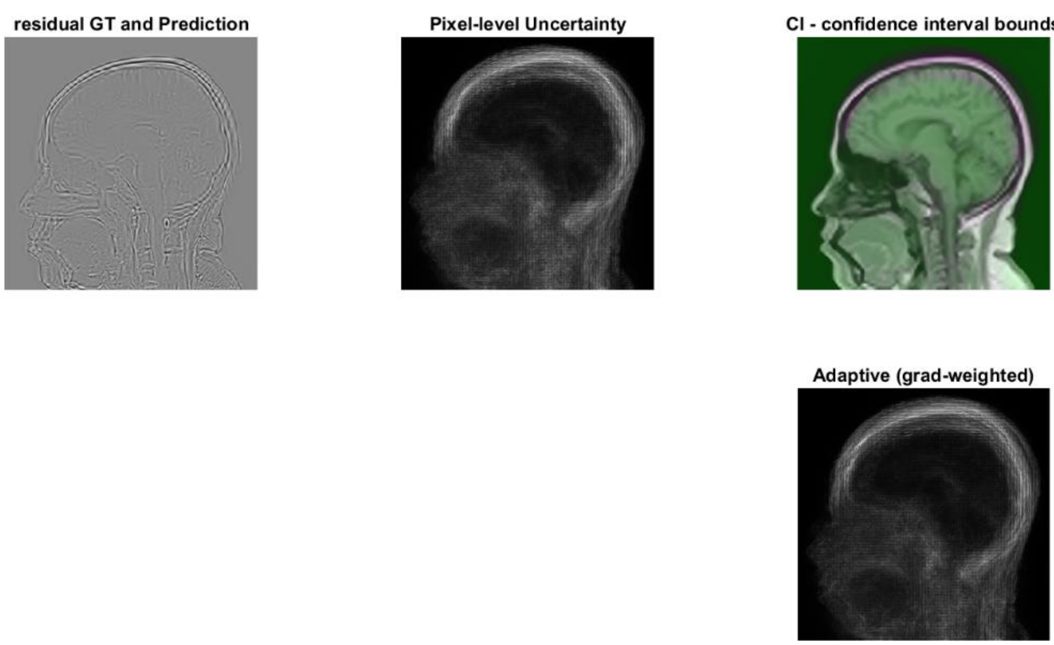

Fig. 3. Outcomes of applying conformal prediction with a = 0.1.

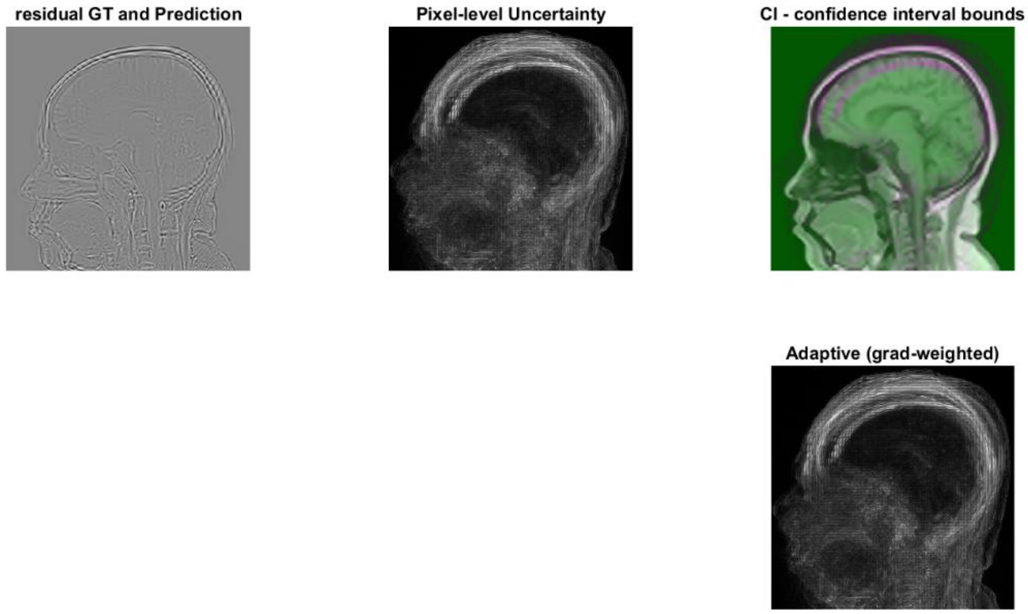

Fig. 4. Outcomes of applying conformal prediction with a = 0.03.

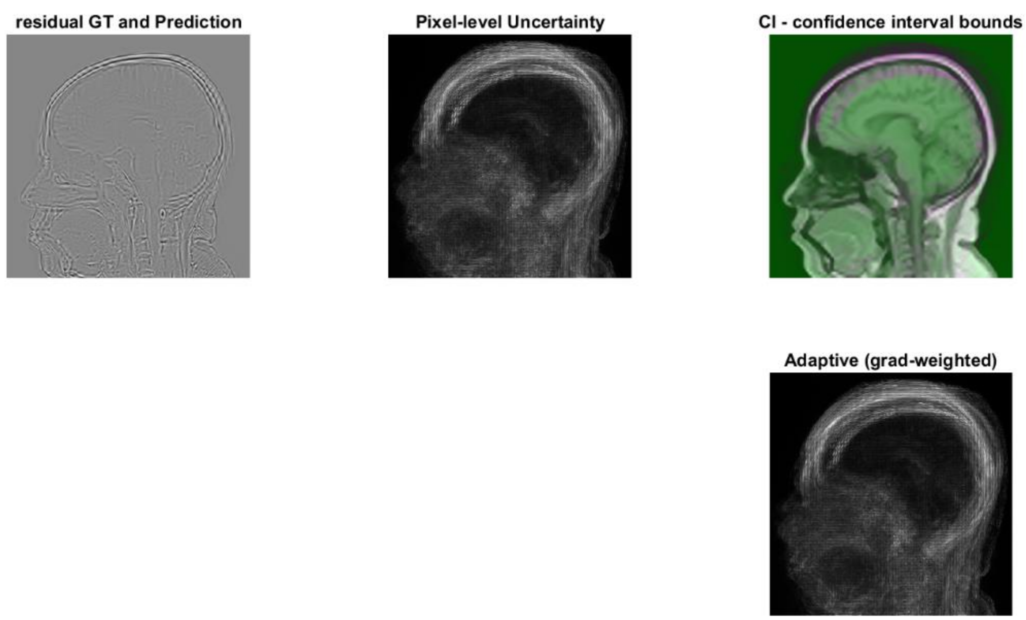

Fig. 5. Outcomes of applying conformal prediction with a = 0.05.

References

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